1.1.3. FINALIZATION PHASE:

I. Abstract for the task: A Use-Case of booking a hotel room: Airbnb:

- Introduction:

This project illustrates a large schema of a database designed for an Airbnb-like platform where its users can rent or book accommodations. To store hosts, guests, bookings, payments, reviews, ratings, locations and supporting features, the ERD documents the primary entities, their attributes, and how they relate to one another. A thorough synopsis of the operations performed by the database administration features, shown in the ERD are provided below.

- Entities and Their Roles:

1. Host, Property and Guest Management:

- Host: Hosts are owners or managers of properties presenting accommodations on rent.They can add properties with photos, prices, and descriptions; handle bookings; respond to guests; take care of properties; enhance lodging experience based on ratings and reviews; have the ability to save payment methods, utilise them to draw payouts and adhere to the guidelines mentioned in the policy and rules. Personal details [name, email, phone number, profile picture, address], payment information, listings uploaded, coupons, gift cards, tutorial and information guides, reviews and ratings, messages to guests, and is-a-verified-host are all stored in the host table.
- Guest: Customer or group of people wanting to rent out accommodations for given time-duration are guests. They can locate listings, make reservations for lodging, communicate with hosts via the messaging feature, provide feedback by rating and reviewing their stay, save appropriate payment methods and use them to make payments, and abide by the mentioned guidelines in the policy and rules. The guest table stores messages with hosts, is-a-verified-guest, payment information, booking history, wishlists, coupons, giftcards, tutorial and information guides, reviews and ratings, and personal information [name, email, phone number, profile photo, and address].
- Listing: Accommodations that hosts offer for guests to rent for a specific amount of time are called listings. The listing table, serving as the main component of the platform, contains information on the property [name, rental photos, description, address, amenities, category], pricing, availability calendar, is-a-guest-favorite, guest reviews and ratings, and is-a-verified-listing.

2. Booking and Transaction Management:

- Booking: Keeps a track of each reservation made for a listing by the guests and confirmed by the host. It also stores details about payments made using attributes[booking date, booking status, payment status]
- Payments: Manages payment processing for guests.
- Payouts: Manages payment processing for hosts.
- Payment Method: Manages payment processing for different types of payment methods used to pay or receive payouts.
- Saved Payment Methods: Manages payment or payout methods saved by the user for convenience.

3. Review and Rating System:

- Review: Allows guests and hosts to offer feedback based on their stay.
- Rating: Measures user experience and enables them to rate guests, accommodations and hosts across several categories.
- Rating-Category: Allows for multiple categories[cleanliness, communication, check-in, accuracy, value and location] for a comprehensive rating framework.
- Rating-Number: Stores and defines the numeric values for ratings to give an in-depth analysis of the given rating.

4. Policy and Rule Enforcement:

- Policy: Stores and mentions platform and listing-specific details[cancellation policy, refund policy] that must be agreed by hosts and guests for a seamless experience
- Rule: Stores and provides a detailed set of instructions[house rules, health and cleanliness rules] for guests and hosts ensuring transparency and compliance.

5. Essentials for Guests and Hosts:
- Amenities: Stores and lists the specific features a listing provides to the guests to enhance their experience and helps in filtering according to the selected amenities.
- Amenities-Category: Provides a catalog of all the available property features
- Listing Category: Provides a list of features guests fancy[mountain view, lake view] while booking a listing.
- Guide Books for Guests: Enables guests to know the tips and tricks to enhance their experience during a stay.
- Guide books for Hosts:Enables host to know the tips and tricks to enhance guests experience during their stay and helps them in being a super host.
- Guide Books Published by Host: Enables hosts to share local tips and guides, improving guest experience.

6. Administrative Oversight:
- Admin: Admins are either managers or owners of the Airbnb platform who ensure rules and agreements are upheld, conflict is resolved, and the system keeps running. Platform usage, maintaining compliance with regulatory needs and community guidelines, conflict resolution regarding cancellations, fines, and refunds are the activities monitored by the admins. Access to different areas and tasks occur based on designation. All the personal details (name, email, phone number, profile picture, and designation) are stored in the admin table.

7. Moderation and Communication:
- Cancellation: Keeps records of cancelled bookings, cancellation reason and refund eligibility for support and resolution.
- Refund: Handles refund requests and processing with regards to cancelled bookings and the payments received.
- Message: Enables communication between guest and host and stores message content.

8. Data Integrity and Relationships:
- Foreign Keys: Foreign keys are used by the tables to connect to the related records[listings -> hosts, bookings -> guests] for maintaining data consistency.
- Many-to-Many Relationships: Enforced via the associative tables[amenities -> listings, rating number -> rating] that provide flexibility to refer data.
- Hierarchical Data: Location, neighbourhood, city, country and region are connected in a hierarchical pattern to provide geographic referring for listings.
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- Conclusion:

This project offers an adequate foundation for an effective platform for renting out properties by balancing user management, booking operations, payment processing, and additional amenities such as location, specified features on a property and reviews and ratings provided by guests for a property. The schema accommodates functional requirements as well as user experience for scalability, data integrity, and functionality. While administrative features ensures efficient governance of the platform, the relationships and structures ensure that there is a seamless integration of all the key operations, ranging from publishing a property to completing a stay and providing feedback

II. Database Metadata Summary:

- Database Name: airbnb_project

A. Database Volume Notes:

1. Total number of tables:

```
                List of relations
 Schema |          Name            | Type  |  Owner
--------+--------------------------+-------+----------
 public | admin                    | table | postgres
 public | amenities                | table | postgres
 public | amenities_category       | table | postgres
 public | booking                  | table | postgres
 public | cancellation             | table | postgres
 public | city                     | table | postgres
 public | country                  | table | postgres
 public | coupon                   | table | postgres
 public | giftcard                 | table | postgres
 public | guest                    | table | postgres
 public | guide_book_for_guest     | table | postgres
 public | guide_book_for_host      | table | postgres
 public | guide_published_by_host  | table | postgres
 public | host                     | table | postgres
 public | id_category              | table | postgres
 public | id_verification          | table | postgres
 public | listing                  | table | postgres
 public | listing_avail            | table | postgres
 public | listing_category         | table | postgres
 public | location                 | table | postgres
 public | msg                      | table | postgres
 public | neighborhood             | table | postgres
 public | pay_category             | table | postgres
 public | payments                 | table | postgres
 public | payouts                  | table | postgres
 public | policy                   | table | postgres
 public | rating                   | table | postgres
 public | rating_category          | table | postgres
 public | rating_number            | table | postgres
 public | receipt                  | table | postgres
 public | refund                   | table | postgres
 public | region                   | table | postgres
 public | review                   | table | postgres
 public | rules                    | table | postgres
 public | saved_payment_methods    | table | postgres
 public | wishlists                | table | postgres
(36 rows)
```

1.

2. Total Database Size:

	total_db_size text
1	9747 kB

3. Table List with Column Count:

Table Name	Columns	Primary Key	Foreign Keys
host	19	host_id	15
guest	19	guest_id	15
listing	16	listing_id	7
rating	6	-	3
review	5	-	2
region	2	region_id	-
country	3	country_id	1
city	3	city_id	1
location	3	location_id	1
neighborhood	3	neighborhood_id	1
payouts	8	payout_id	1
payments	6	payment_id	1
msg	3	-	-
guide book for host	4	guide_host_id	-
guide book for guest	4	guide_guest_id	-
policy	3	policy_id	-
rules	3	rules_id	-
coupon	4	coupon_id	-
giftcard	4	giftcard_id	-
refund	8	-	4
cancellation	9	-	4
booking	12	booking_id	2
receipt	15	receipt_id	3
listing category	2	listing_category_id	-
wishlists	4	wishlist_id	2
id verification	2	id_verify	-
listing avail	8	listing_id	1
saved payment methods	3	saved_pay_id	2
admin	27	admin_id	13
rating category	2	rating_category_id	-

Table Name	Columns	Primary Key	Foreign Keys
guide published by host	5	published_guide_id	1
amenities	5	amenities_id	3
amenities categories	3	-	-
rating number	2	-	-
pay category	2	-	-
id category	2	-	-

4. Number of Entries[count per table, as of schema]

Table Name	Entries [Count per table]
host	20
guest	20
listing	20
rating	20
review	20
region	6
country	55
city	78
location	80
neighborhood	79
payouts	40
payments	40
msg	92
guide book for host	19
guide book for guest	14
policy	12
rules	7
coupon	20
giftcard	20
refund	20
cancellation	20
booking	40

Table Name	Entries [Count per table]
receipt	40
listing category	20
wishlists	20
id verification	60
listing avail	20
saved payment methods	40
admin	20
rating category	6
guide published by host	20
amenities	20
amenities categories	75
rating number	51
pay category	6
id category	9

5. Total Number of Commits and Rollbacks performed on database:

	db_commit bigint	db_rollback bigint
1	3338	44

B. Data Observation Notes:

- Schema Factors:

1. Highly normalised structure with extensive foreign key relationships
2. Usage of constraints for data integrity maintenance. Constraints used are:
   - Primary Keys - Explicitly defined
   - Foreign Keys - Use Match Simple With No Action
   - Not Null - Implemented on critical columns

3. Specialised data types are used for a comprehensive database schema. Below mentions all the data types used for this schema:

Data type	Description
Character varying	For fields that include characters as well numeric information such as ID, address, email and passwords

Data type	Description
Character	For fields that includes characters only such as name or different category[eg. listing-cateory]
Date	For fields that includes date only such as booking date, payment date and guests since
Bytea	For fields that stores images/ image location in binary format
Numeric	For fields that includes numbers only such as phone-number or different category[eg. rating-number]
Boolean	For fields that includes flags [True/False] only such as is-guest-verified, is-a-superhost, is-a-guest-favourite
Double precision	For fields that includes precise numerics only such as total_amt, nightly_price, discounted_amt
Text	For fields that includes longer descriptions such as listing-desciption, policy/rules content, guidebooks
Time without time-zone	For fields that includes timestamp without zone specifications such as message_sent
Date range	For fields that includes a range of date such as coupon validity, giftcard validity